

The empirical recurrence for OEIS A203721, proved by transfer matrix

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Abstract

OEIS A203721 counts arrays of width 2 over $\{0, 1, 2\}$ through two derived arrays of PAIR SUMS — each cell added to its left neighbour, and each cell added to the one above it — on which the entry imposes an order condition. The entry carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: the count is a walk count on $S = 10$ states.

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1 The conjecture

OEIS A203721 is “Number of $(n+1) \times 2$ arrays with column and row pair sums $b(i,j)=a(i,j)+a(i,j-1)$ and $c(i,j)=a(i,j)+a(i-1,j)$ nondecreasing in column and row directions, respectively.”. It has offset 1 and begins

32, 113, 326, 888, 2278, 5653, 13665, 32440, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 6*a(n-1) - 11*a(n-2) + a(n-3) + 16*a(n-4) - 10*a(n-5) - 7*a(n-6) + 6*a(n-7) + a(n-8) - a(n-9)$.

As of the “Last modified” line on the live entry (Jun 04 2018 14:17:42, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array a has 2 cells across, with entries in $\{0, 1, 2\}$. Two derived arrays are formed,

$$b(i, j) = a(i, j) + a(i, j - 1) \quad (j \geq 1), \quad c(i, j) = a(i, j) + a(i - 1, j) \quad (i \geq 1),$$

so b adds each cell to its LEFT neighbour and c adds each cell to the one ABOVE it. The entry asks

$$b(i, j) \leq b(i+1, j) \quad \text{and} \quad c(i, j) \leq c(i, j+1)$$

wherever both sides are defined: the array b must not decrease DOWN a column and the array c must not decrease ALONG a row. The two directions are crossed, which is the only subtle part of the reading — b is built across a row and tested down a column, and c the other way about.

3 Two rows decide it

Lemma 1. *Both conditions are conditions on two consecutive rows of a and nothing more.*

Proof. $b(i, j) \leq b(i+1, j)$ names four cells of a , all in rows i and $i+1$. $c(i, j) \leq c(i, j+1)$ names four cells, all in rows $i-1$ and i . $\square$

So a single row of a is a state and one step appends a row.

With M the adjacency matrix of the digraph on those states, ι the indicator of the starting state and $\tau = \mathbf{1}$,

$$a(n) = \iota^\top M^j \tau, \quad S = 10.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = 6a(n-1) - 11a(n-2) + a(n-3) + 16a(n-4) - 10a(n-5) - 7a(n-6) + 6a(n-7) + a(n-8) - a(n-9)$$

for every $n > 8$.

Proof. The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 8 + 1$ onwards and the run continued until the working vector was identically zero. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 26 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size, form the two derived arrays in full, and test the condition on them directly. No window, no state and no incremental bookkeeping are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.